

Bench grinding machine

4.4 Precision Grinding Machine

Precision grinders are used for finishing parts to a very accurate dimension. The main types of precision grinders are

1. Cylindrical grinding machines
2. Internal grinding machines
3. Surface grinding machines
4. Tool and cutter grinding machines
5. Special grinding machines

Cylindrical grinding machine

The workpiece is held between two centres and used to grind external surfaces like cylinders, taper cylinders and cylindrical faces is called cylindrical grinding machine. There are two types of cylindrical grinding machines, they are

1. External cylindrical grinding machine
2. Internal cylindrical grinding machine

External cylindrical machine

In a cylindrical work piece to grind external surface like cylindrical and taper cylinders is called external cylindrical grinding machine.

Base

The base is made of cast iron and rest on the floor. It supports all other parts of the grinder. The top of the base is accurately machined and provides guideways for the table to slide on. The base houses the table driving mechanism.

Tables

The tables are mounted on top of the base. There are two tables namely upper table and lower table. The lower table slides on the guideways on the bed. Table can be moved by hand or power.

Upper table is mounted on the lower table. Upper table can be swiveled upto ±10° and clamped in position. Adjustable trip dogs are clamped in longitudinal slots at the side of the lower bed. These trip dogs actuate the table reversing lever, to control table movement.

Head stocks

The head stock is situated at the leftside of the upper table. It supports the workpiece by means of a centre and drives it by means of a dog. It may hold and drive the workpiece in a chuck. It houses the mechanism meant for driving the work. The head stock of a universal grinding machine can be swiveled to any required angle.

Tailstock

The tailstock is situated at the rightside of the table. It can be adjusted and clamped in various positions to accommodate different lengths of workpieces.

Wheel head

The wheel head is placed over the bed at its backside. The wheel head may be moved at right angles to the table guideways. It is operated by hand or by power to feed the wheel to work. The wheel head carries a grinding wheel.

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